The empirical recurrence for OEIS A318041, recovered from its tail

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Abstract

OEIS A318041 records "Empirical recurrence of order 72" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 100$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 3$, so the recurrence is proved for every $n \geq 75$. Its last coefficient is $c_{72} = -16 \neq 0$, so no further tail satisfies anything shorter; any order-72 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A318041 is "Number of nX4 0..1 arrays with every element unequal to 0, 2, 3 or 4 horizontally, vertically or antidiagonally adjacent elements, with upper left element zero.". It has offset 1 and begins

$$1, 3, 21, 140, 905, 6179, 42793, 297043, \ \ldots$$

The entry states:

> Empirical recurrence of order 72 (see link above)

As of the "Last modified" line on the live entry (Aug 13 2018 14:31:20, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \ = \ \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 100$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 100$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. ☐

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 72: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 3$ is the first whose minimal order is exactly the stated 72.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 200$ exact terms from $n = 3$ on, it returns order 72 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{72} c_i \, a(n-i),$$

c_1	15	c_{19}	−42531	c_{37}	−3620756	c_{55}	−3759587
c_2	−89	c_{20}	28181	c_{38}	−4868774	c_{56}	−1812407
c_3	329	c_{21}	−245981	c_{39}	−951047	c_{57}	2278306
c_4	−899	c_{22}	−220411	c_{40}	4075573	c_{58}	3651083
c_5	1724	c_{23}	1118404	c_{41}	4309616	c_{59}	1170899
c_6	−2333	c_{24}	476574	c_{42}	5884083	c_{60}	−854303
c_7	2287	c_{25}	1246900	c_{43}	3036695	c_{61}	−807652
c_8	−737	c_{26}	572004	c_{44}	−8929143	c_{62}	70239
c_9	757	c_{27}	−2691797	c_{45}	−17479547	c_{63}	302392
c_{10}	−3816	c_{28}	−4807245	c_{46}	−17639021	c_{64}	86492
c_{11}	4203	c_{29}	−4852073	c_{47}	−10182822	c_{65}	−67808
c_{12}	−9283	c_{30}	−5590696	c_{48}	2829200	c_{66}	−76573
c_{13}	−10707	c_{31}	505838	c_{49}	7915766	c_{67}	−23743
c_{14}	36355	c_{32}	6870188	c_{50}	7322577	c_{68}	−3854
c_{15}	−3507	c_{33}	10664622	c_{51}	5147909	c_{69}	1497
c_{16}	−18445	c_{34}	13475783	c_{52}	1277478	c_{70}	340
c_{17}	125324	c_{35}	9581322	c_{53}	−916079	c_{71}	96
c_{18}	−141457	c_{36}	−251193	c_{54}	−3184718	c_{72}	−16

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 75$, and it is the recurrence OEIS A318041 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 3$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{72} - c_1 x^{72-1} - \cdots - c_{72}$. Its constant term is $-c_{72} = 16$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$

vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 72 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 72, so $\deg q = 72$, and it is monic. It holds from some index t on. From $\max(t, 3)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 72, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 21 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 72, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -16.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-72 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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